Megan B. Bourque

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Education

University of Tennessee Knoxville, Knoxville, TN *08/2020 – 12/2024*

B.A. Physics, Minor in Geology – GPA 3.36/4.0

Chattanooga State Community College, Kimball, TN *05/2018- 08/2021*

No Degree - GPA 3.878/4.0

- Dual enrollment with high school and University of Tennessee Knoxville.

Skills

- ArcGIS Pro
- QGIS
- Google Suite
- Microsoft Excel, PowerPoint, and Word
- Python and MATLAB

Experience

Muse Knoxville 02/2025 – Present

STEAM Outreach Educator, Mobile Planetarium Lead

- Traveled across East Tennessee to deliver engaging, hands-on STEAM programs to Pre-K through 5th grade students in schools, libraries, and community centers.
- Delivered presentations to audiences ranging from toddlers to adults, developing strong science communication skills and the ability to make technical content approachable and exciting.
- Operated and presented shows using a Mobile Planetarium, tailoring astronomy content to suit a wide range of age groups and educational backgrounds.
- Designed and adapted educational content to align with learning standards.
- Trained new educators and volunteers.

University of Tennessee Knoxville - Department of Earth, Environmental, and Planetary Science *12/2022 - 12/2024*

Undergraduate Research Assistant

- Used monostatic, bistatic, and stereo Mini-RF data using ArcGIS to analyze and create maps of the lunar surface.
- Became proficient with ESRI ArcGIS Pro and to collect data and create hazard maps for research on potential landing sites for NASA's Artemis mission.
- Presented this research at the international Lunar and Planetary Science Conference and local science conferences.

University of Tennessee Knoxville - Department of Physics & Astronomy *01/2022 – 12/2024*

Undergraduate Teaching Assistant

- Assisted students to better understand the fundamentals of physics and astronomy.
- Trained graduate Teaching Assistants on how to use ground-based telescopes owned by the University of Tennessee Knoxville
- Used ground-based telescopes to teach undergraduate students about astronomy, often providing supplementary information to what students have learned from lectures.
- Aided in leading supplementary lectures for students as a review for quizzes and exams.
- Collaborated closely with a professor to organize class materials and grade students' submitted works.
- Wrote and edited weekly newsletters with supplementary information to aid students with lessons learned in class.

NASA & ASU L'SPACE Mission Concept Academy *01/2022 – 05/2022*

- This is a 12-week competitive course to improve skills in mission development by training each week with NASA scientists and engineers.
- Worked with a team to design a space mission under guidance of NASA employees, resulting in a 90-page Preliminary Design Review.
- Earned the following skill badges: Organizational Charts, Gantt Charts, Requirements, Risk Management, Systems, Heat Systems, Budgets

Presentations

55th Lunar and Planetary Science Conference, The Woodlands, TX *March 2024*

Hazard Mapping of the Southern Lunar Surface: Candidate Artemis Landing Site - Malapert Massif

EUReCA 2024, Knoxville, TN *April 2024*

Hazard Mapping of the Southern Lunar Surface: Candidate Artemis Landing Site - Malapert Massif

Awards & Honors

Volunteer of Distinction Award from the University of Tennessee Knoxville – March 27, 2025

NASA Space Grant Undergraduate Award – May 2, 2024

The Otto Kopp Undergraduate Research Award – May 2, 2024

Tennessee NASA Space Grant - Summer 2023, Fall 2023, Spring 2024

Advanced Undergraduate Research Activity (AURA) – Summer 2023

Faculty Research Assistant FRAF at the University of Tennessee Knoxville – Spring 2023

Leadership Experience

- Astronomy Club at the University of Tennessee Knoxville – President for 2 years, Vice President for 1 year, Secretary for 1 semester
- Society of Physics Students - Secretary for 1 year, Historian for 1 year, Outreach Coordinator for 1 year